SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Total Marks: 20

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Code of Conduct

I Rakshana. A. R (192126002 7) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R. R

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Cloud Computing Fundamentals:

*Cloud computing is a model that provides computing resources and services over the Internet whenever they are required. Instead of purchasing & maintaing all the computing hardware & software locally, users can access resources such as servers, storage, application & processing power through a cloud provider.

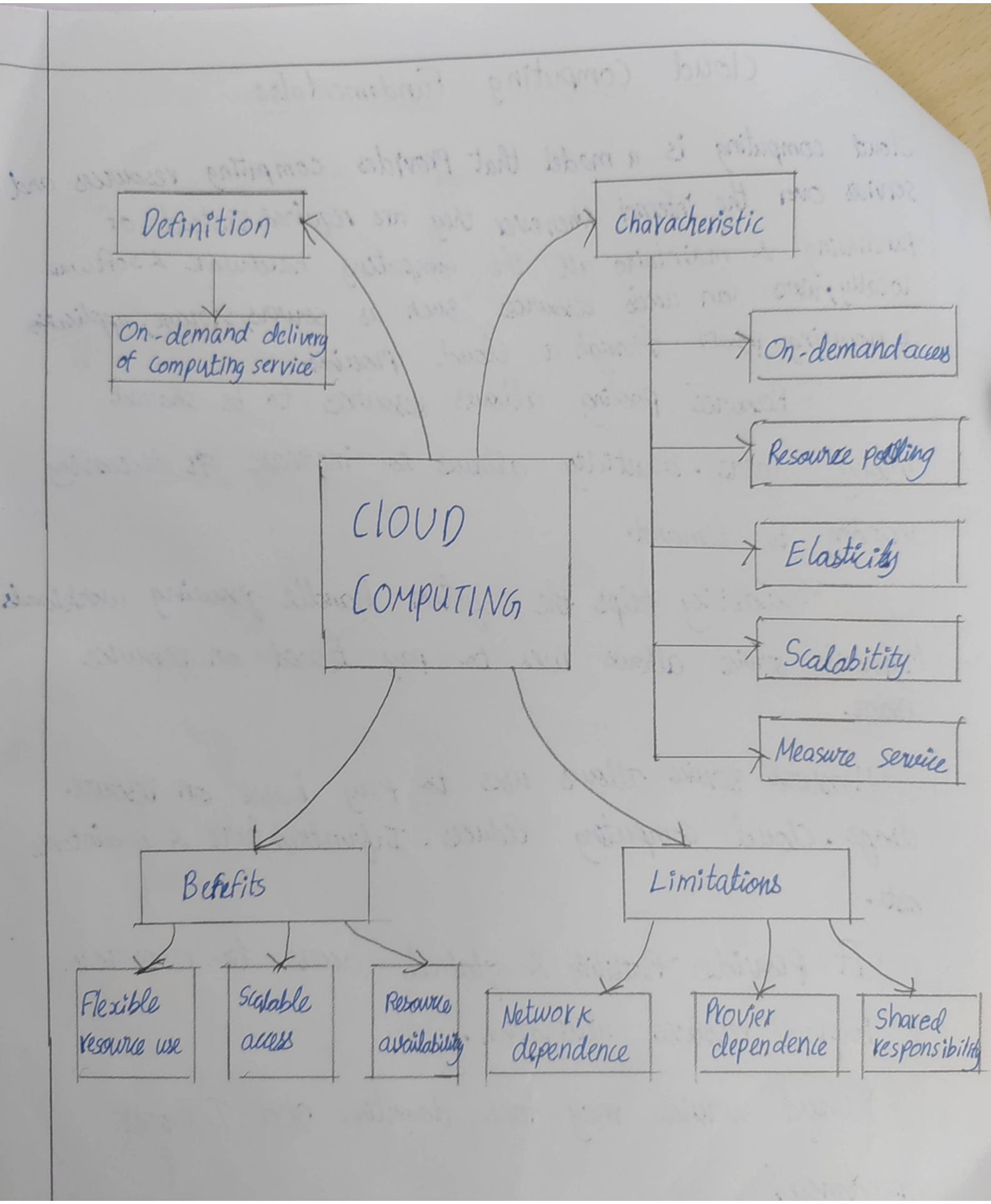
* Resource pooling allows resources to be shared among users. Elasticity allows to increase or decreasing according to demand.

*Scalability helps the system handle growing workloads. Measured service allows uses to pay based on resource usage.

*Measured service allows uses to pay based on resoure usage. Cloud computing reduces Infrastructure & maintance costs.

*It provides flexibility & global auess to resources & improves resource utilization.

* Cloud service may face downtime and Internet dependancy.


CLOUD COMPUTING
Definition
On-demand delivery of computing service
Characteristic
On-demand access
Resource pooling
Elasticity
Scalability
Measure service
Benefits
Flexible resource use
Scalable access
Resource availability
Limitations
Network dependence
Provier dependence
Shared responsibility

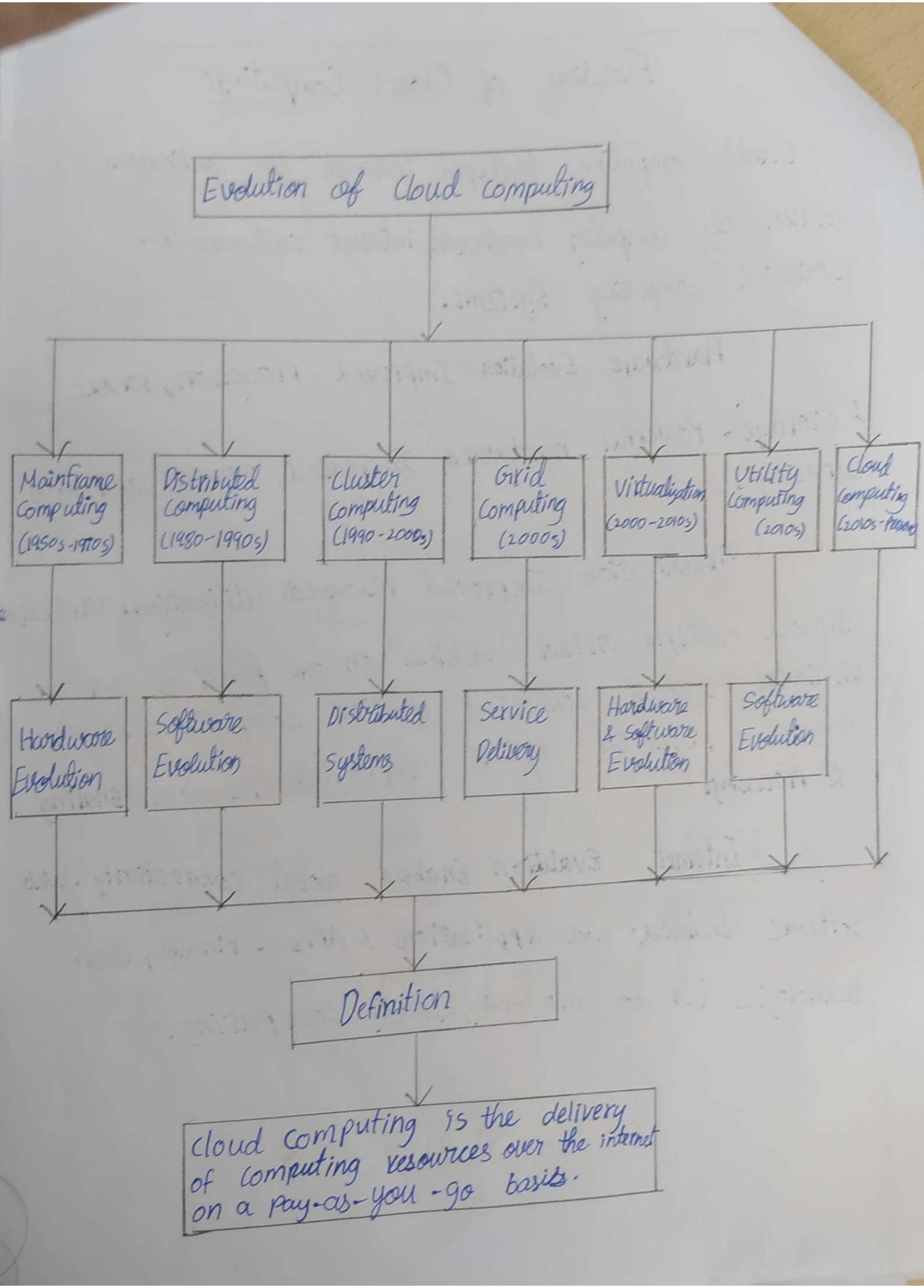
Evolution of Cloud Computing:

* Clould computing developed through the continuous evolution of computer hardware, internet software & distributed computing systems.

* Hardware Evolution Improved processors, servers & storage. Powerful hardware Increased overall computing capacity.

* Virtualization Improved resoused utilization. Virtualization allowed multiple virtual Machines on one physical server. Distributed systems Connected multiple computer through networks. Distributed systems supported resource sharing & processing.

* Internet Evalution Enabled global connectivity. Web Software Enabled web applications & APIs. Finally, these technologies led to modern cloud computing.


Evolution of Cloud computing
Mainframe Computing (1950s-1970s)
Distributed Computing (1980-1990s)
Cluster Computing (1990-2000s)
Grid Computing (2000s)
Virtualization (2000-2010s)
Utility Computing (2010s)
Cloud computing (2010s-Present)
Hardware Evolution
Software Evolution
Distributed Systems
Service Delivery
Hardware & Software Evolution
Software Evolution
Definition
Cloud Computing is the delivery of computing resources over the internet on a pay-as-you-go basis.

Service Models:

* Cloud service models describe the different levels of computing resources and serveices provided to users through the cloud. The differ according to what the provider manages and what the customer is responsible for.

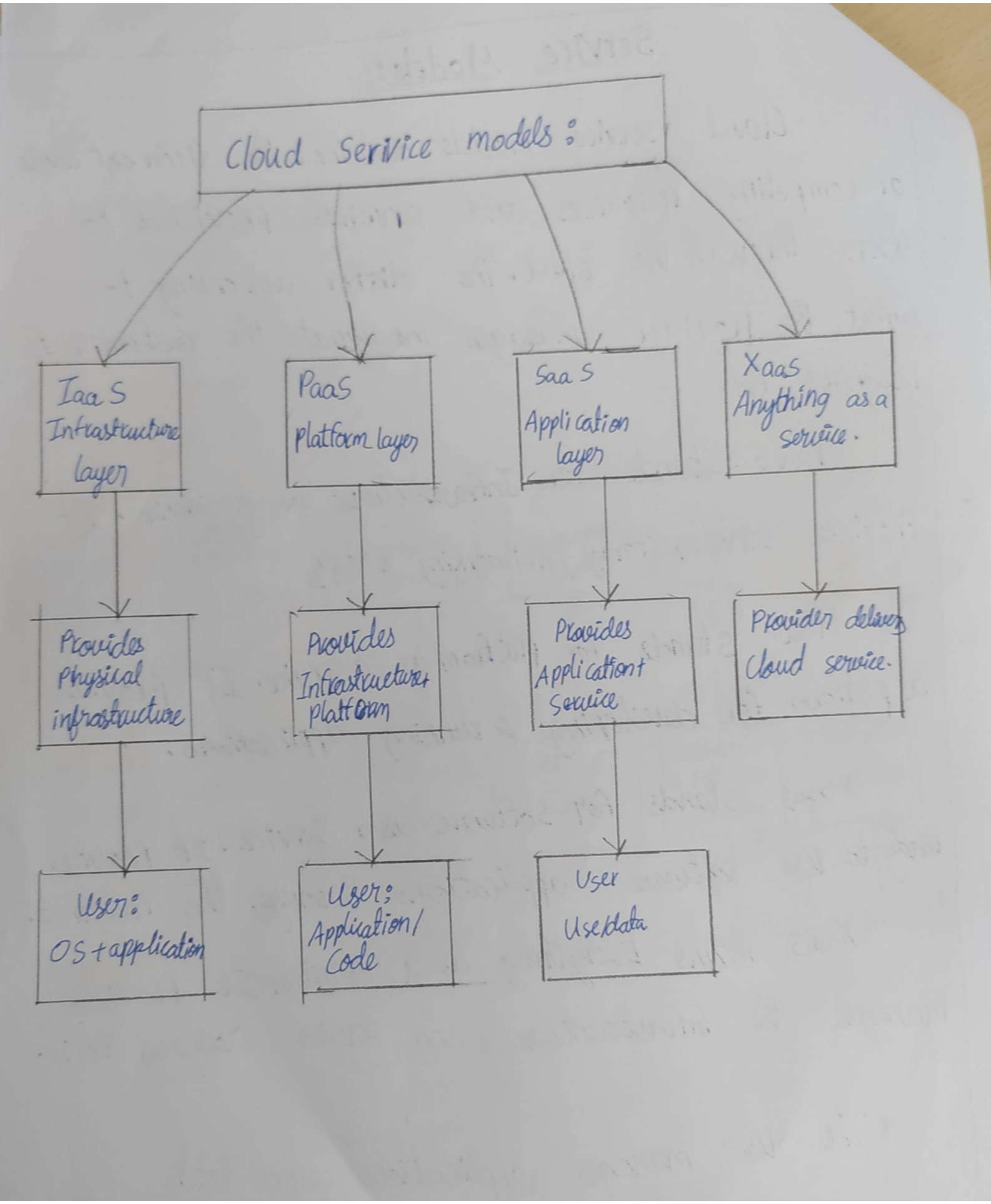
* Iaas Stands for Infrastructure as a Servie. It Provides serves, Storage, networking & VM's

* Paas stands for platfrom as a service. It provides a platform for developing & running applications.

* Saas stands for Software as a Service. It Provides ready to use software applications through the Internet.

* Xaas means Everything as a service. It Provides manages the Infrastructure and servios assigned to it.

* The user manages applications and data according to the service model.


Cloud Service models:
IaaS
Infrastructure layer
PaaS
platform layer
SaaS
Application layer
XaaS
Anything as a service.
Provides Physical infrastructure
Provides Infrastructure+ platform
Provides Application Service
Provider delivers Cloud service.
User: OS + application
User: Application/ Code
User
User data

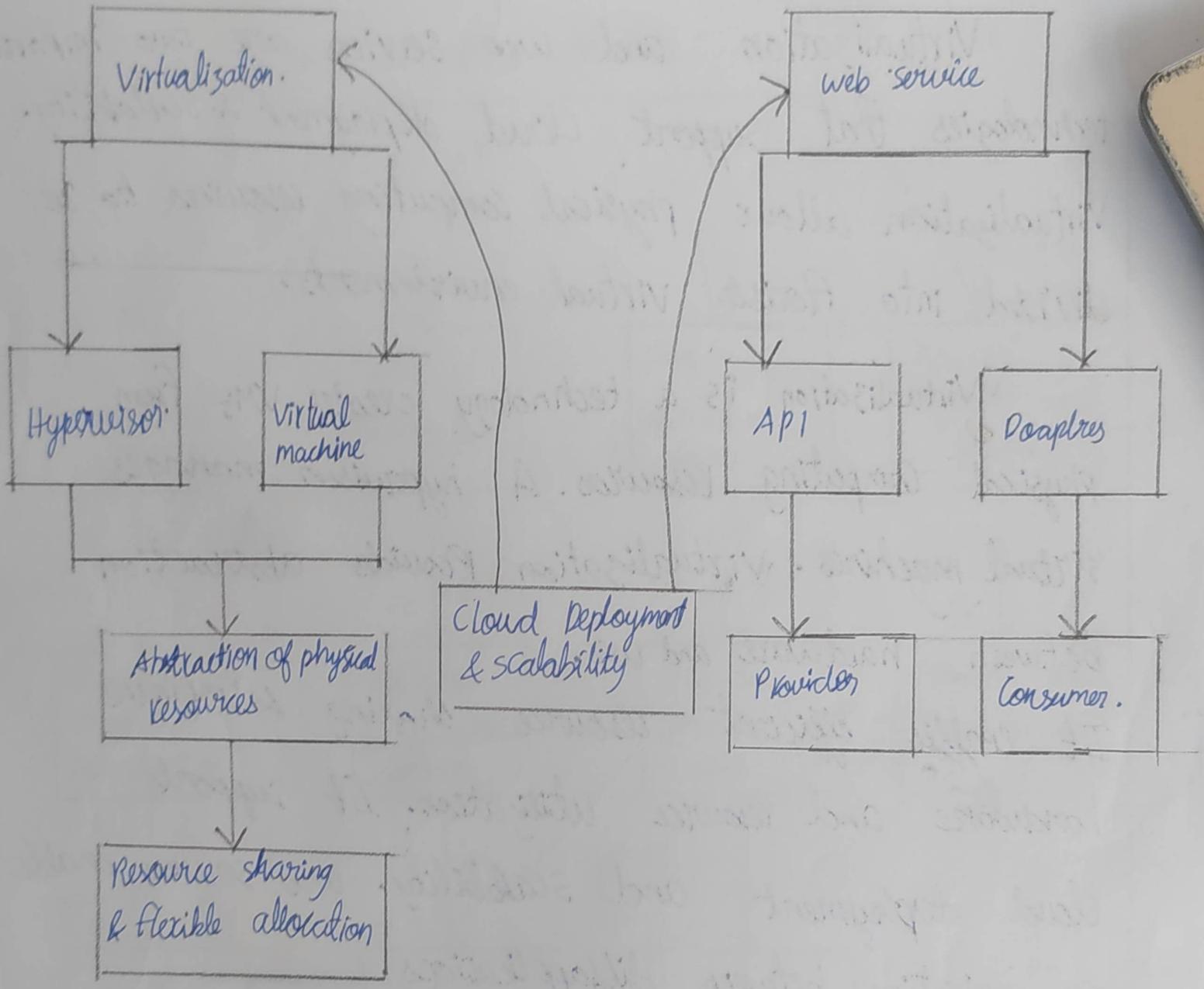
Virtualization & Web services as Enables of Cloud:

*Virtualization and web services are two important technologies that support cloud deployment & scalability. Virtualization allows physical computing resources to be divided into flexible virtual environments.

*Virtualization is a technology creates VMs from physical computing resources. A hypervisor manages virtual machines. Virtualization Provides abstraction between hardware and uses.
It enables efficient resource sharing & improves hardware and resource utilization. It supports cloud deployment and scalability. web services enable communication between diffapplications.

*APIs Provide access to cloud services and resources. SOAP and REST all used for web-based communication.

*Web services support interopesability between di Systems.


Virtualization.
Hypervisor.
Virtual machine
Abstraction of physical resources
Resource sharing & flexible allocation
Cloud Deployment & scalability
web service
API
Doaplres
Provider
Consumer.